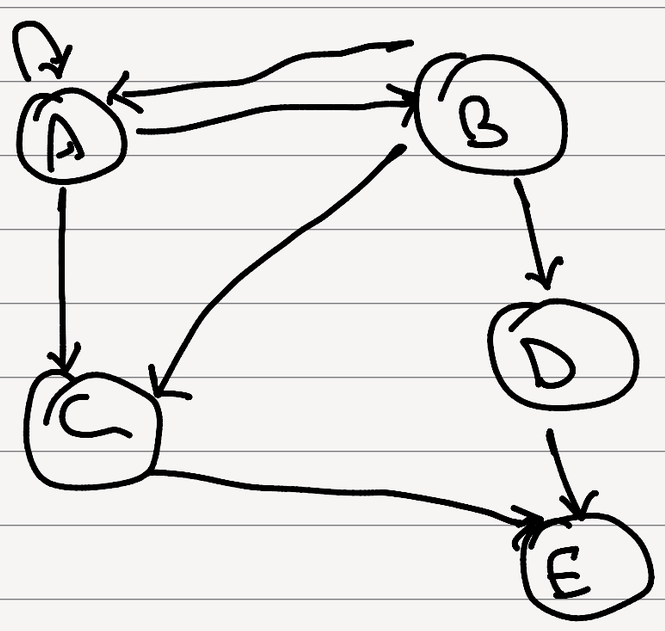
Esercizi sul PageRank

1)



$r_j = \sum_{i \to j} r_i / d_i$ → Link uscenti

$r_A = \frac{r_A}{3} + \frac{r_B}{3}$

$r_B = \frac{r_A}{3} +$

$r_C = \frac{r_A}{3} + \frac{r_B}{3}$

$r_D = \frac{r_B}{3}$

$r_E = r_C + r_D$

$r_A + r_B + r_C + r_D + r_E = 1$

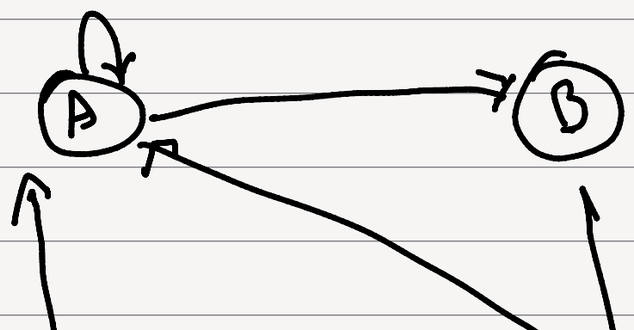
Matrice:

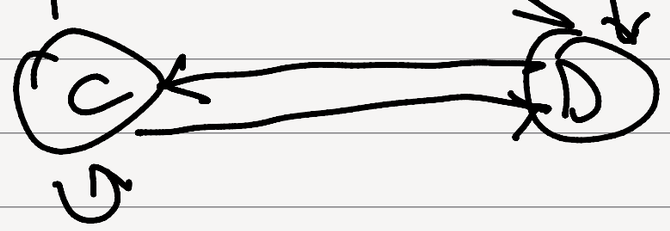
	A	B	C	D	E
A	1/3	1/3	0	0	0
B	1/3	0	0	0	0
C	1/3	1/3	0	0	0
D	0	1/3	0	0	0
E	0	0	1	1	0

La matrice non è stocastica perché la somma delle colonne E non è 1 in quanto E è un ded end. Trasformiamola in stocastica:

β (A B C D E ; A 1/3 1/3 0 0 0 ; B 1/3 0 0 0 0 ; C 1/3 1/3 0 0 0 ; D 0 1/3 0 0 0 ; E 0 0 1 1 0) $+ (1-\beta)$ (A B C D E ; A 1/5 1/5 ⋮ ⋯ ⋯ ; B 1/5 ⋯ ⋯ ⋯ ⋮ ; C ⋮ ⋮ ⋮ ⋮ ; D ⋮ ⋮ ⋮ ⋮ ; E ⋮ ⋮ ⋮ 1)

$\sim$ $\sim$





$$r_A = \frac{r_A}{2} + \frac{r_C}{3} + \frac{r_D}{2}$$

$$r_B = \frac{r_A}{2}$$

$$r_C = \frac{r_C}{3} + \frac{r_D}{2}$$

$$r_D = r_B + \frac{r_C}{3}$$

$$r_A + r_B + r_C + r_D = 1$$

$$M = \begin{array}{c} \begin{matrix} A & B & C & D \end{matrix} \\ \begin{pmatrix} 1/2 & 0 & 1/3 & 1/2 \\ 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 1/2 \\ 0 & 1 & 1/3 & 0 \end{pmatrix} \end{array} \begin{matrix} A \\ B \\ C \\ D \end{matrix}$$

M è stocastica in questo caso